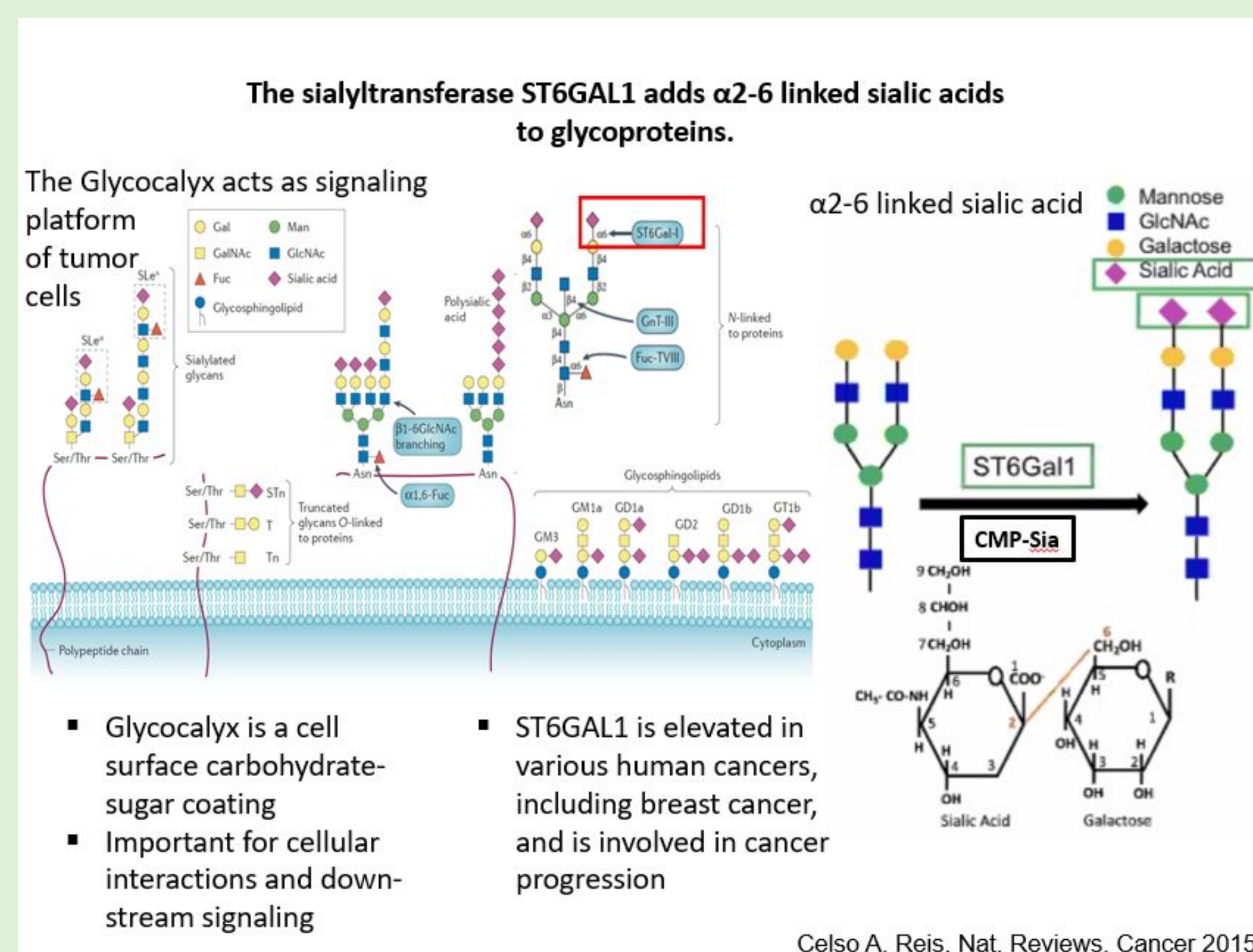


Abstract

Glycosylation is an essential post-translational modification of proteins that dictates cellular functions. The Sialyltransferase ST6GAL1-mediated α 2,6-sialylation of glycoproteins has been implicated in regulating various events, including cell growth and survival, integrin- β 1 function, inflammation, and cancer. ST6GAL1 resides within the ER-Golgi secretory apparatus of cells that produce them and is implicated in mediating its functions. ST6GAL-1 is prominently associated with human cancers, including breast cancers. However, many studies have found that enzymatically active ST6GAL1 is also released into the extracellular space. Extracellular ST6GAL1 has been shown as a potent modifier of hematopoiesis and in the sialylation of the anti-inflammatory IgG. However, the impact of this extrinsic mechanism of ST6GAL1 on tumor cell pathobiology is unknown. In cancers, elevated ST6GAL1 generally predicts poorer patient outcomes, but the reason for these outcomes is unknown. We hypothesize that extracellular ST6GAL1 signaling is essential for aggressive triple-negative breast cancer (TNBC) cell growth and survival. We used recombinant ST6GAL1 (rST6G) for extracellular ST6GAL1 (exoST6G) protein and CMP-Sia, as the enzyme substrate for extracellular glycosylation mechanisms. Human TNBC MDA-MB-231 and BT549 cell models were used to examine extracellular ST6GAL1-mediated cancer cell biology. Our results show that the exoST6G mechanism enhances cancer cell α 2,6 linked sialylation of glycoproteins. ExoST6G probably enhances tumor cell growth and survival by modulating cell-surface growth factor receptors, e.g., epidermal growth factor receptor (EGFR). In conclusion, our data suggested that extracellular ST6GAL1 signaling is important for cancer cell growth, survival, and invasion and is an easy target for therapy, which has strong clinical implications

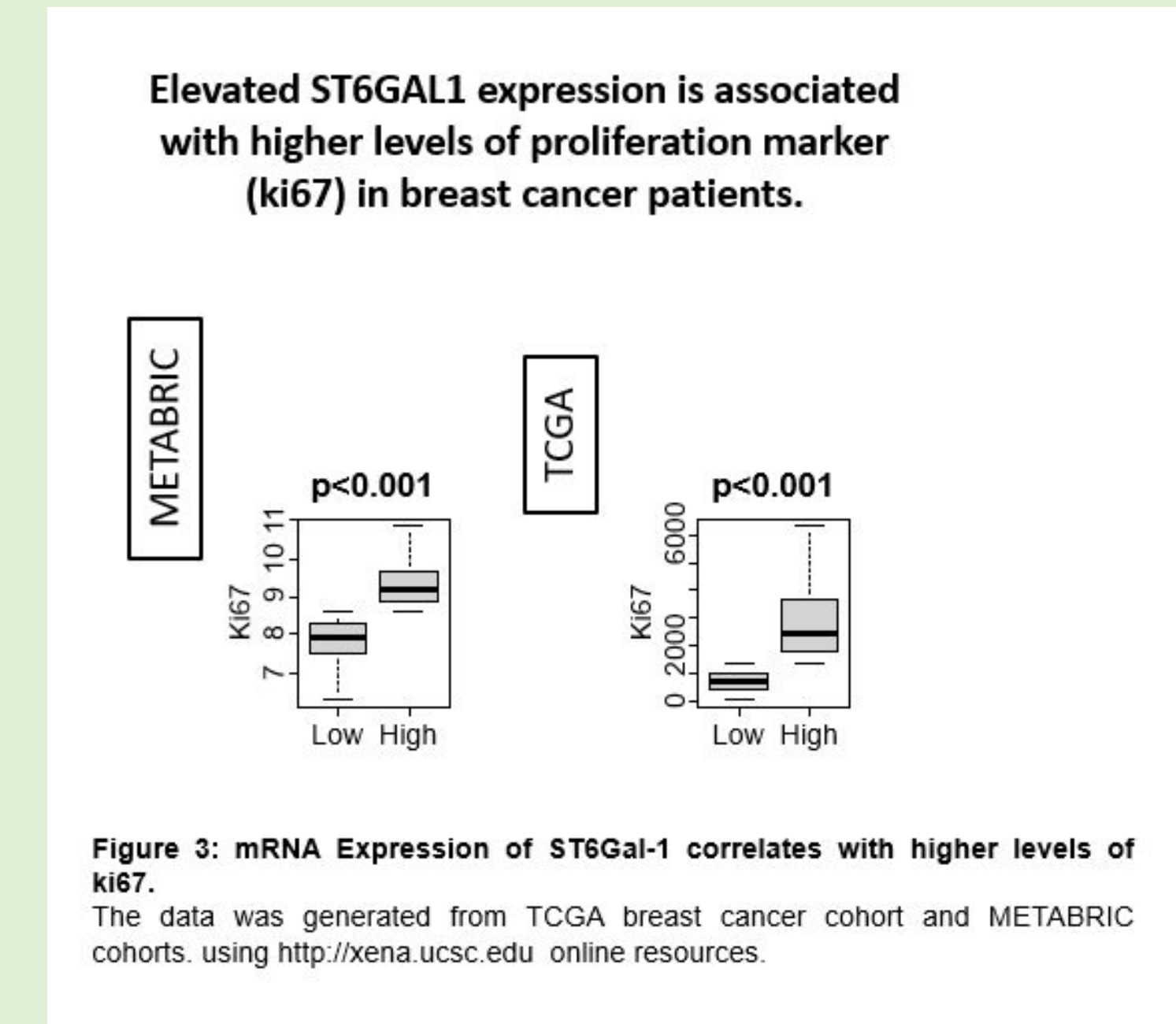
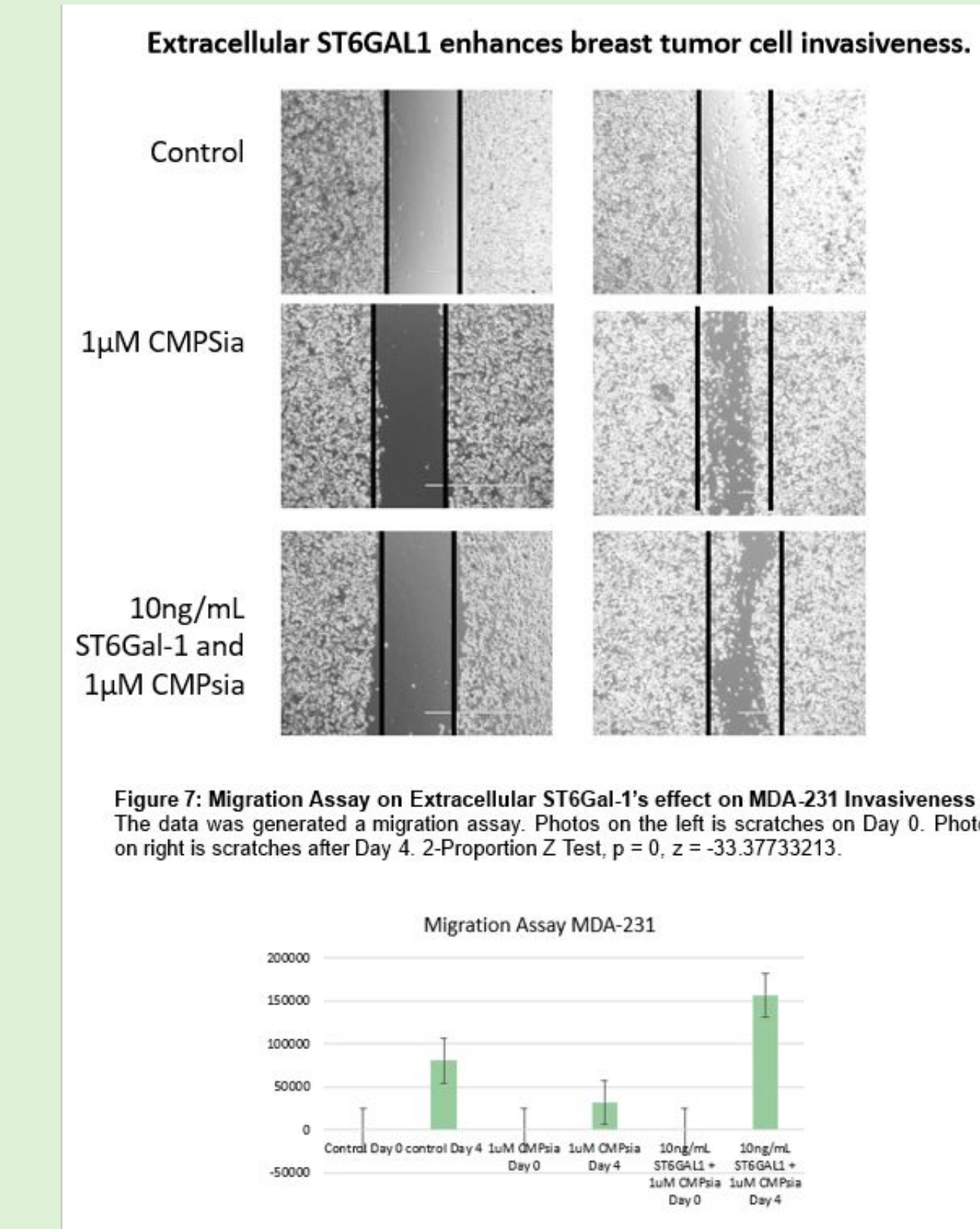
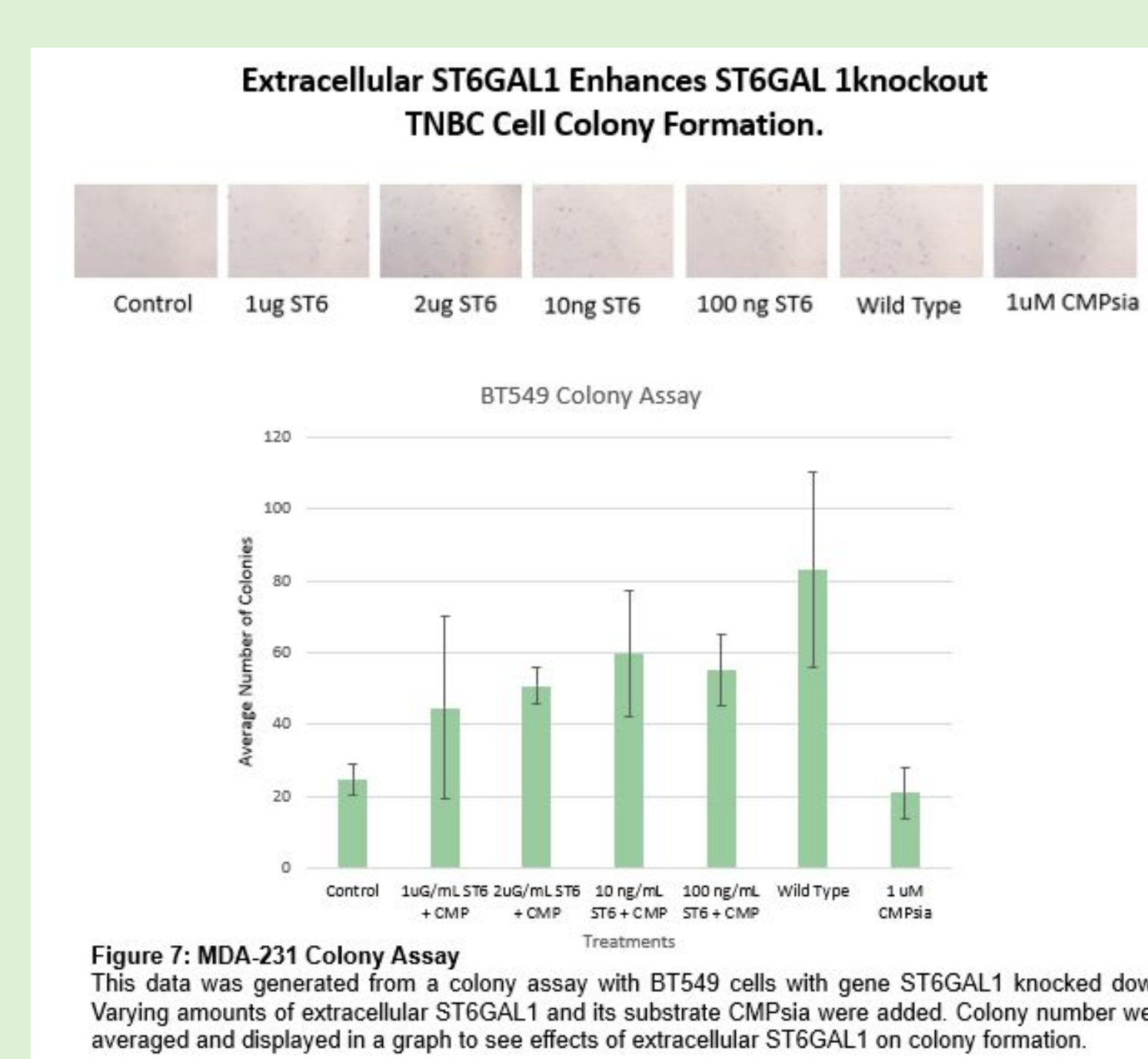
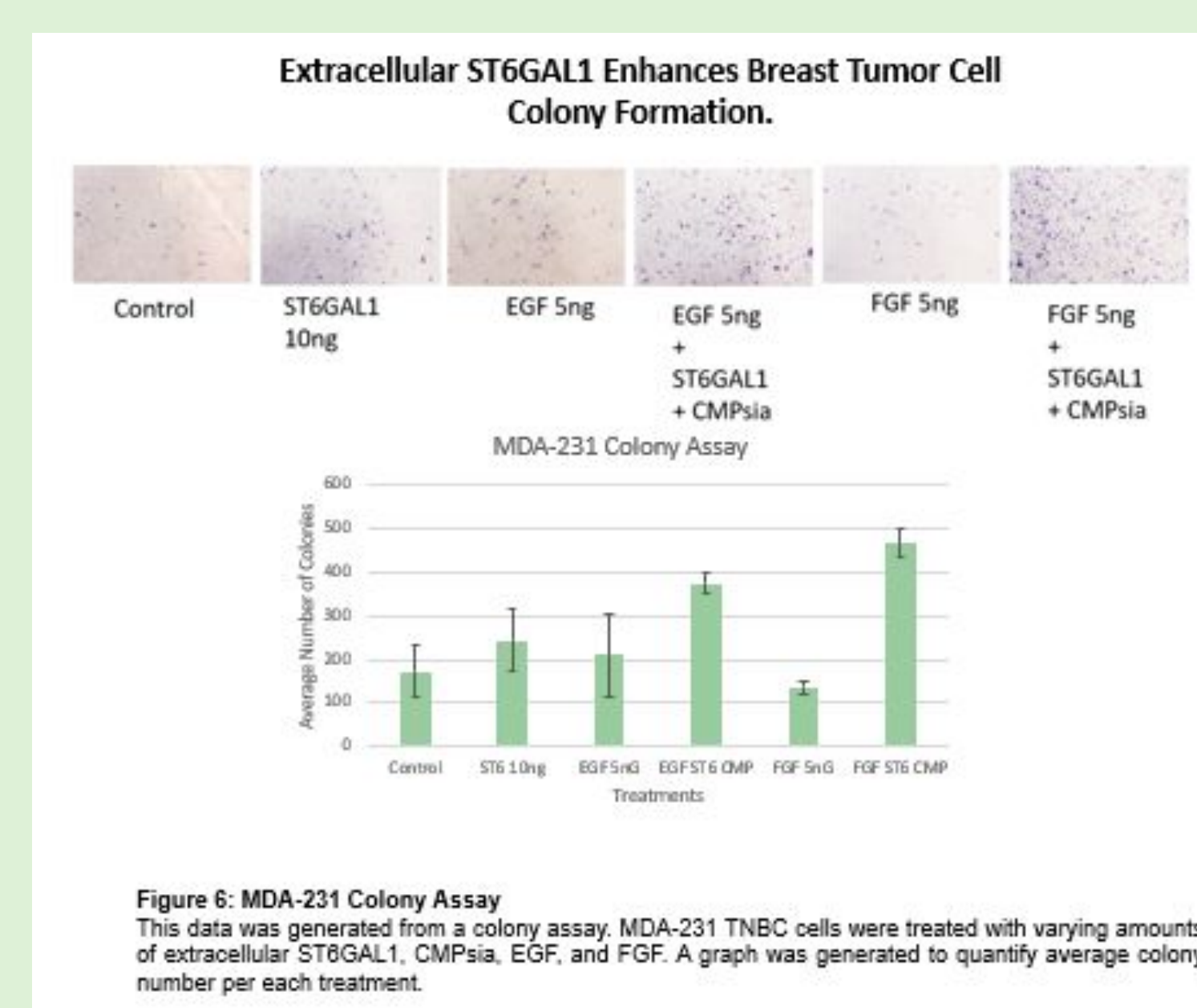
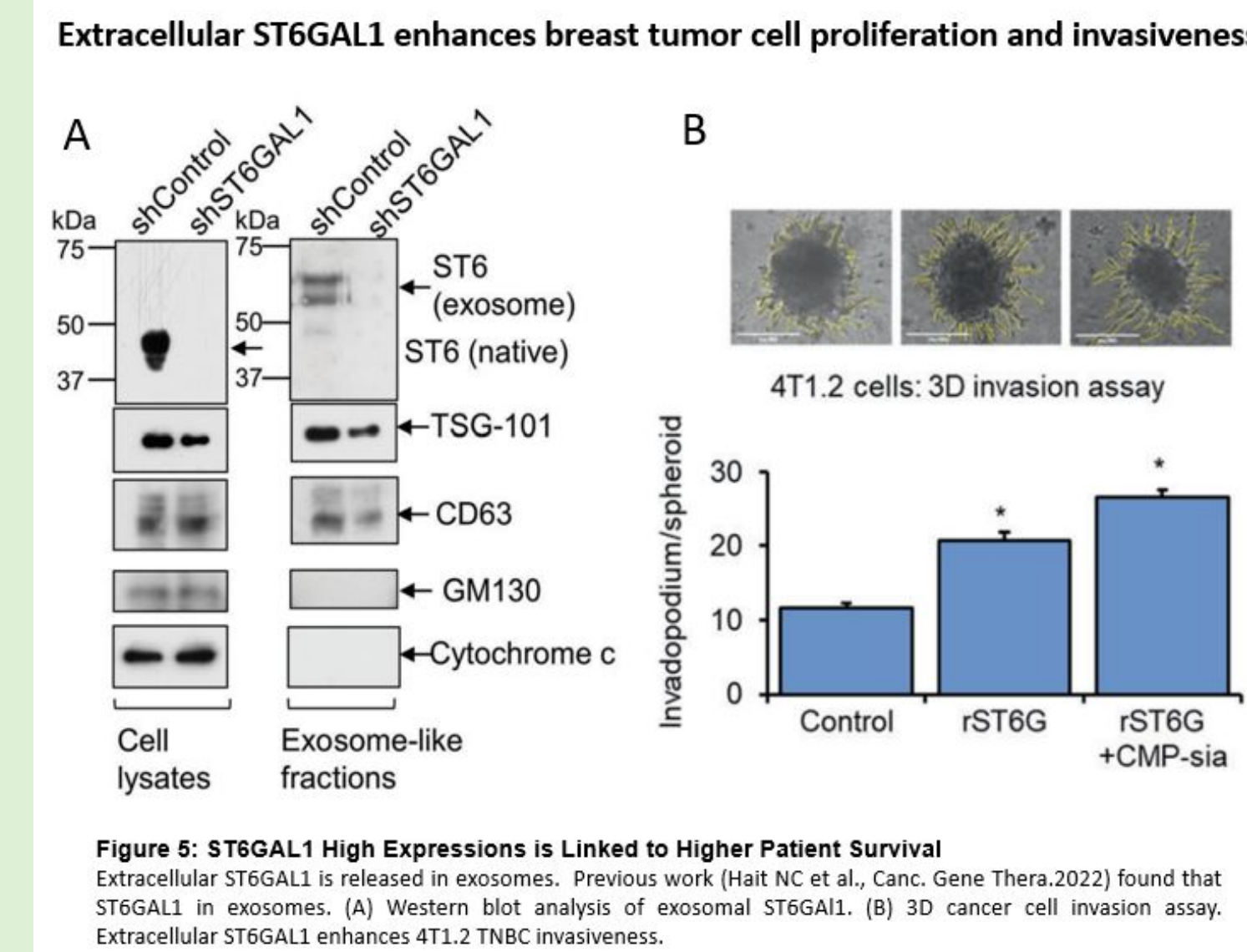
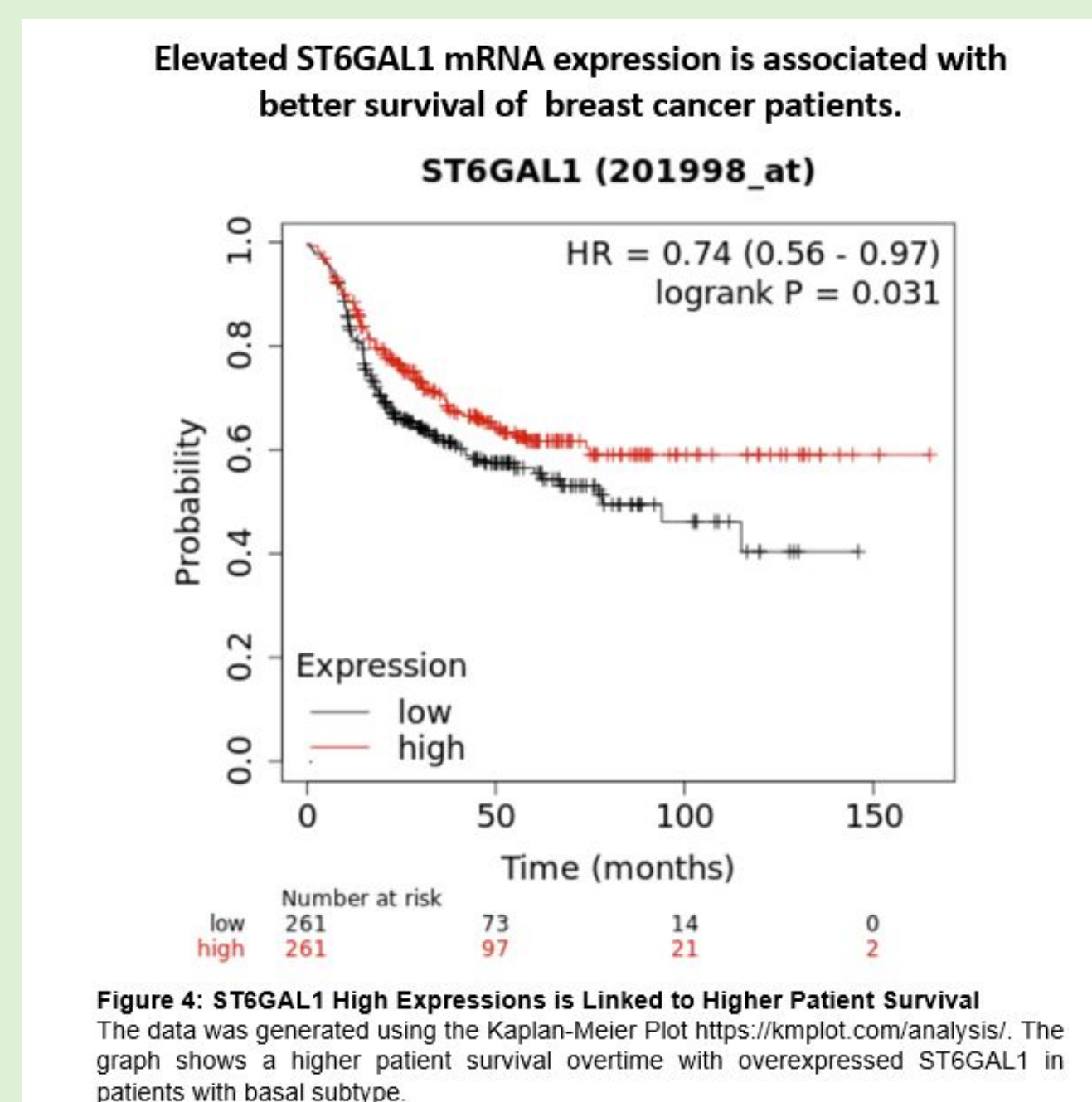
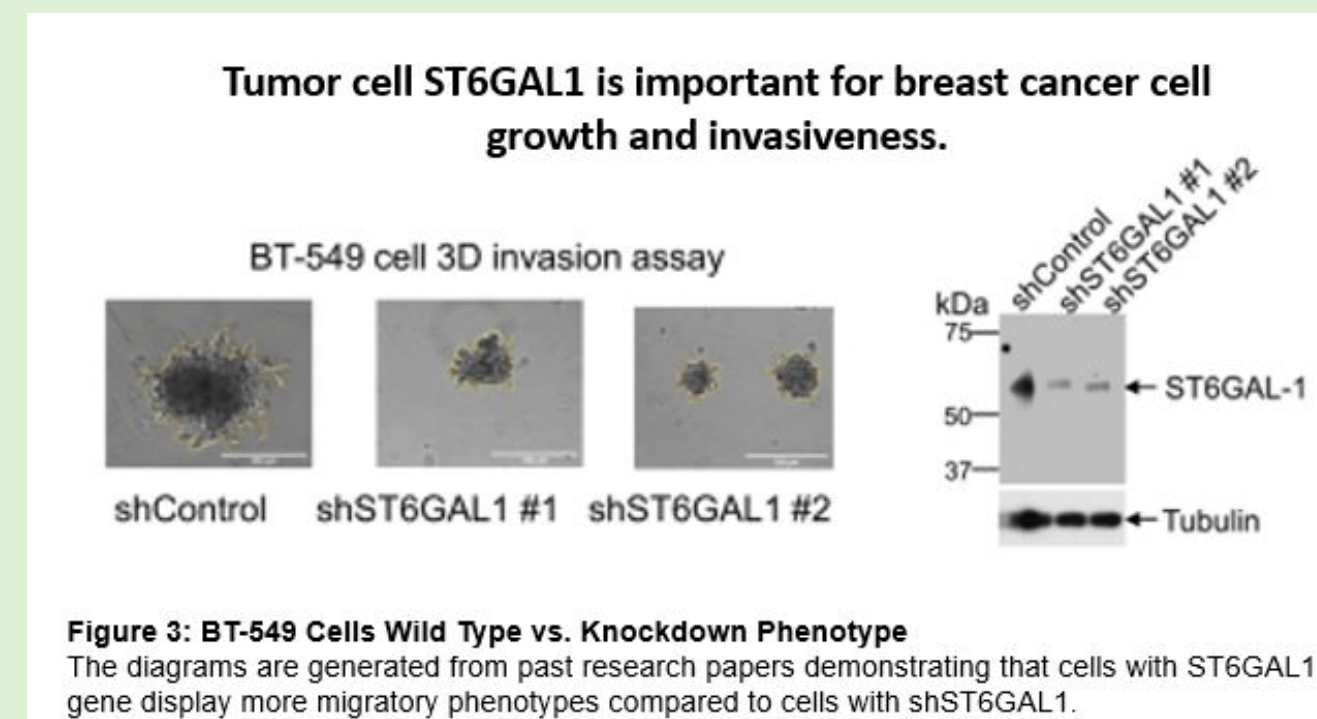
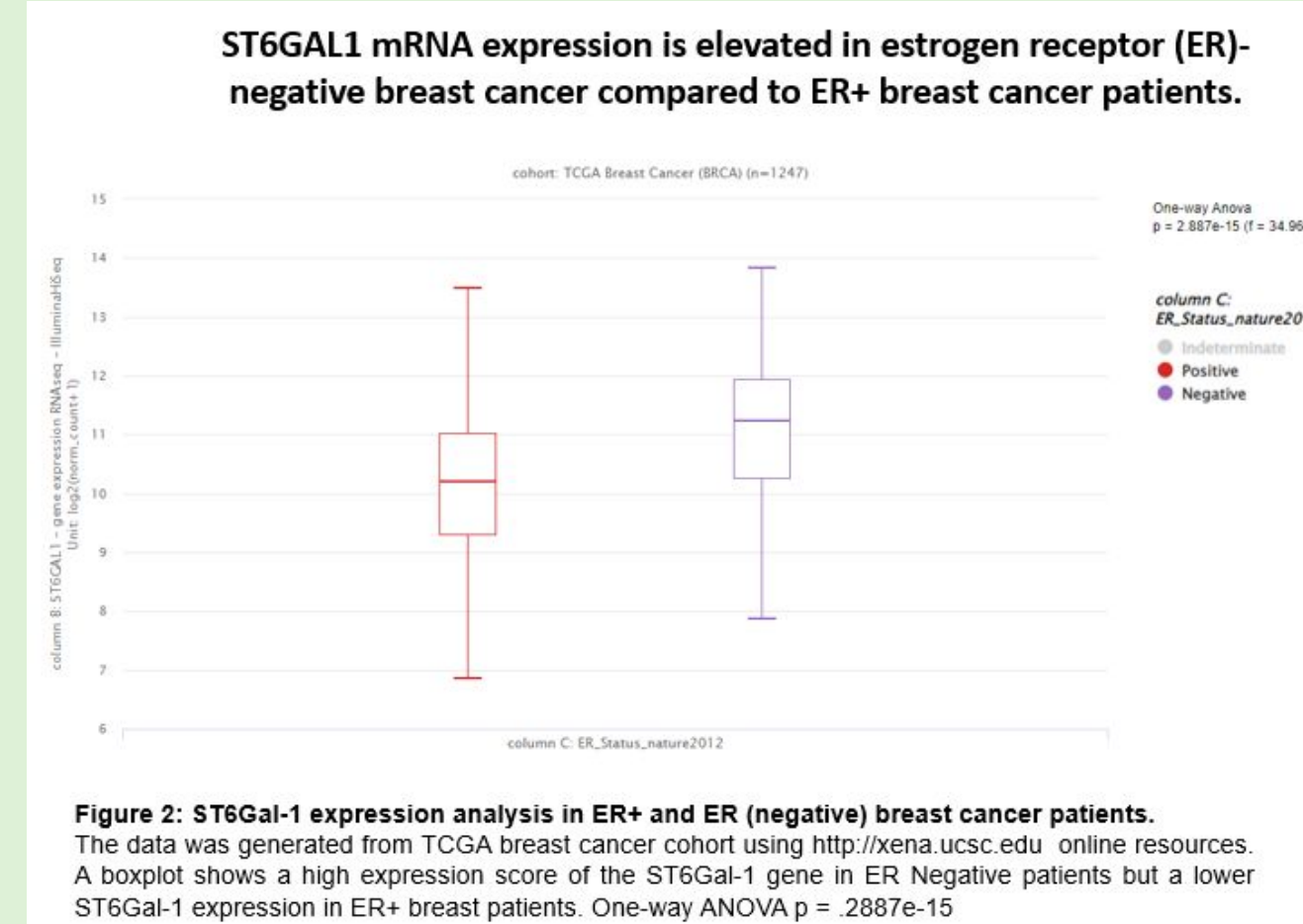
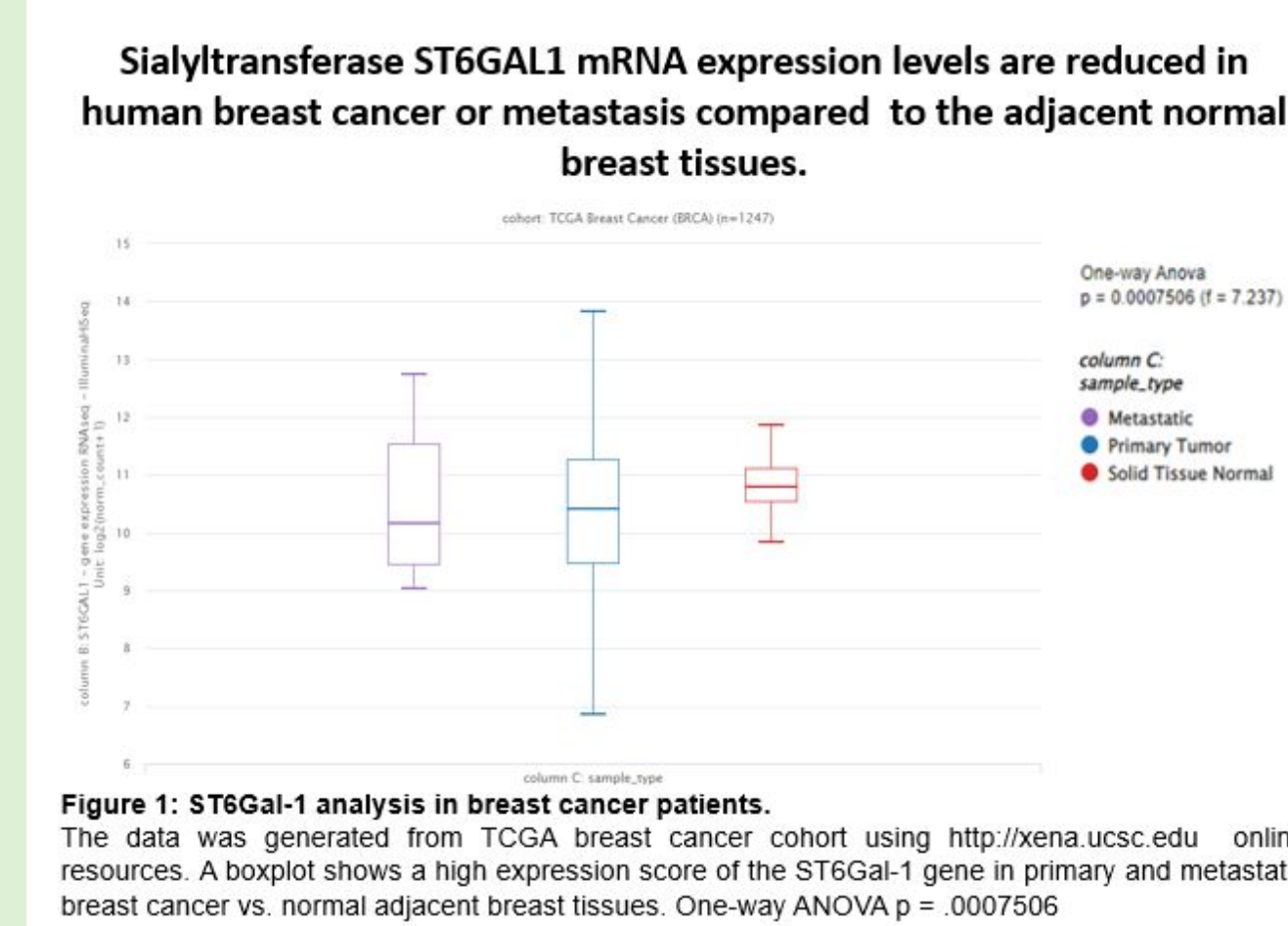
Introduction:



Methods

Wound assay was created in 6-well dish filled with MDA-231 cells. Each well was scratched, treated, and photographed on Day 0 and Day 4. A colony assay were performed using Triple Negative Breast Cancer Cell lines MDA-231. MDA-231 was treated with various amounts of ST6GAL1, its substrate CMPsia, and growth factors EGF and FGF. Another colony assay was performed using BT549, a mouse cell line, with ST6GAL1 knocked down. Colonies were treated with varying amounts of ST6GAL1 and its substrate CMPsia.

RESULTS



CONCLUSION

- Our data appears to support a relationship between high expression of ST6GAL1 and higher levels of cancer cell proliferation.
- Our migration assay also provides data that MDA-231 cancer cells have increased migration speeds when provided with extracellular ST6GAL1 and its substrate CMPsia in vitro.
- However, elevated ST6GAL1 mRNA levels are also found to correlate with higher long term patient survival despite showing more aggressive cancer behavior.
- St6gal1knockout TNBC cells showing metastatic properties with extracellular ST6GAL1 signaling.
- Our data provide evidence that may explain heterogeneous expression of tumor ST6GAL1-linked cancer progression.
- Since extracellular ST6GAL1 can be easily blocked, so our research has strong clinical significance in aggressive breast cancer, e.g., TNBC.

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